

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8873/01

24 September 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, class and tutor's name on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **14** printed pages.

[Turn over

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1** In the discovery of an element, **A**, scientists found that it consists of 3 isotopes, ^{140}A , ^{142}A and ^{144}A in the ratio of 2 : 3 : 1.

What is the relative atomic mass of element **A**?

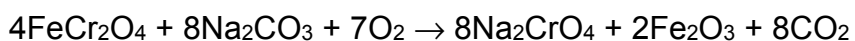
- A** 140.5
- B** 141.7
- C** 142.0
- D** 143.1

- 2** *Use of the Data Booklet is relevant to this question.*

How many Cl atoms are present in 1 cm³ of chlorine gas under room conditions?

- A** $\frac{1 \times 24\,000}{6.02 \times 10^{23}}$
- B** $\frac{1 \times 6.02 \times 10^{23} \times 2}{24\,000}$
- C** $\frac{1 \times 6.02 \times 10^{23} \times 2}{71}$
- D** $\frac{1 \times 6.02 \times 10^{23}}{24}$

- 3** A constituent of wood preservative is manufactured by heating iron(II) chromite, FeCr_2O_4 , with sodium carbonate in air.



Which species is oxidised in this process?

- A** chromium only
- B** oxygen only
- C** chromium and iron
- D** carbon and oxygen

- 4 An isolated gaseous species has paired electrons in at least one of its 3d orbitals and a fully filled 4s subshell.

What could be the identity of the species?

- A Cu
- B Fe^{3+}
- C Ni^{2+}
- D Sr^{2+}

- 5 The elements aluminium and phosphorus each form triply charged ions.

How do the atomic radii and ionic radii of these elements compare?

- A $\text{P}^{3-} > \text{Al} > \text{P} > \text{Al}^{3+}$
- B $\text{P}^{3-} > \text{P} > \text{Al} > \text{Al}^{3+}$
- C $\text{P} > \text{Al} > \text{P}^{3-} > \text{Al}^{3+}$
- D $\text{P} > \text{P}^{3-} > \text{Al}^{3+} > \text{Al}$

- 6 Which one of the following species is **not** planar?

- A BrF_3
- B ICl_4^-
- C PCl_4^+
- D XeF_4

- 7 Hydrazine, N_2H_4 , and hydrogen peroxide, H_2O_2 , are both used as rocket propellants because they can produce large volumes of hot gases from a small volume of liquid.

Which of the following statements about these two compounds are correct?

- 1 The bond angle in N_2H_4 is larger than that in H_2O_2 .
- 2 The N–H bond is shorter than the O–H bond.
- 3 Hydrazine forms stronger intermolecular hydrogen bonds than hydrogen peroxide.
- 4 Both hydrazine and hydrogen peroxide are polar molecules.

- A 1 and 4 only
 B 2 and 3 only
 C 1, 2 and 4 only
 D 1, 3 and 4 only

- 8 Three elements, **E**, **F** and **G**, have the physical properties shown in the table.

element	melting point / °C	boiling point / °C	density / g cm ⁻³
E	–7	59	3.12
F	98	883	0.97
G	649	1107	1.74

What could be the identities of **E**, **F** and **G**?

	E	F	G
A	Br_2	Al	Si
B	Br_2	Na	Mg
C	I_2	Mg	Na
D	I_2	Si	K

- 9 Which statement about exothermic reactions is **incorrect**?
- A On the reaction pathway diagram, the products of the reaction are lower than the reactants.
 - B There is a net transfer of heat energy from the reacting system to the surroundings.
 - C The total bond energies of the products > the total bond energies of the reactants
 - D The activation energy for the forward reaction is the same as that for the backward reaction.
- 10 A student mixed 25.0 cm³ of 3.00 mol dm⁻³ hydrochloric acid with an equal volume of 6.00 mol dm⁻³ sodium hydroxide. The initial temperature of both solutions was 15.0 °C. The maximum temperature recorded was 24.5 °C. It was found that 15 % of the heat produced during the experiment was lost to the surroundings.

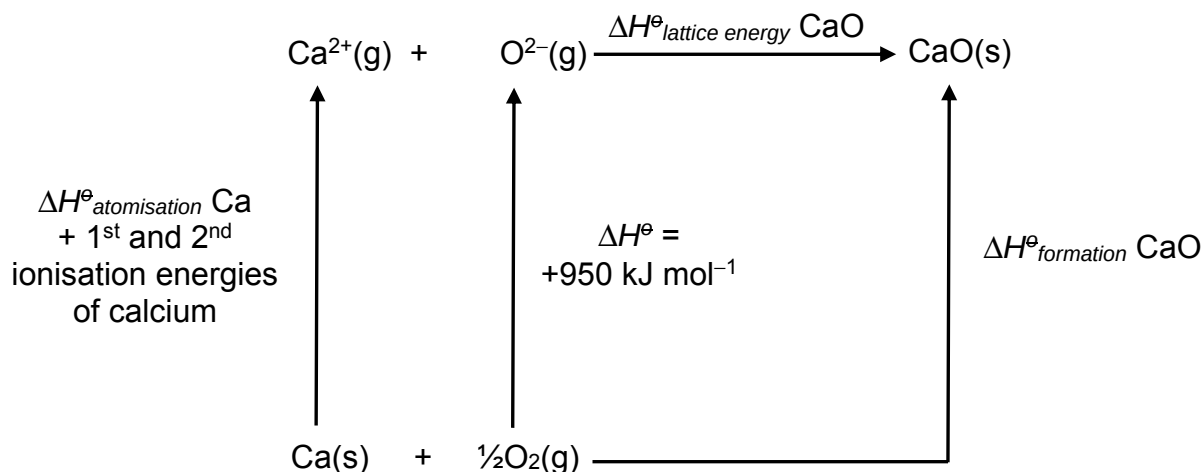
Using these results, what is the enthalpy change of neutralisation?

- A -15.6 kJ mol⁻¹
- B -22.5 kJ mol⁻¹
- C -31.1 kJ mol⁻¹
- D -57.3 kJ mol⁻¹

[Turn over

- 11 A student calculated the lattice energy for calcium oxide using the cycle shown.

The enthalpy change of atomisation of calcium, $\Delta H^\ominus_{\text{atomisation}} \text{Ca}$, is the energy needed when 1 mol of gaseous calcium atoms is formed from 1 mol of solid calcium under standard conditions.



However, the value calculated by the student for the lattice energy was **more** exothermic than the correct value.

Which errors could have been made in the calculation?

- A omitting the value of $+950 \text{ kJ mol}^{-1}$
- B omitting the standard enthalpy change of formation of calcium oxide
- C using the 1st and 2nd ionisation energies of magnesium instead of calcium
- D using the standard enthalpy change of combustion of calcium rather than the standard enthalpy change of formation of calcium oxide

- 12 The graph below shows the Boltzmann distribution of molecular energies at a given temperature.



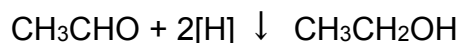
As temperature increases, which statements are correct?

- 1 The maximum of the curve is displaced to the right.
- 2 The maximum of the curve is displaced downwards.
- 3 The proportion of molecules with energies **above** any given value increases.
- 4 The proportion of molecules **with** any given energy increases.

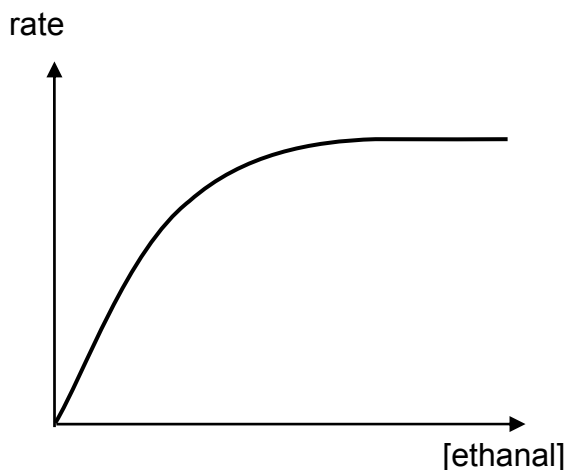
- A 1 only
B 3 only
C 1, 2 and 3 only
D 1, 2, 3 and 4

[Turn over

- 13 The enzyme *alcohol dehydrogenase* catalyses an important step in the production of ethanol by fermentation.



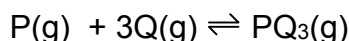
The graph shows how the rate of this enzyme-catalysed reaction varies with the concentration of ethanal.



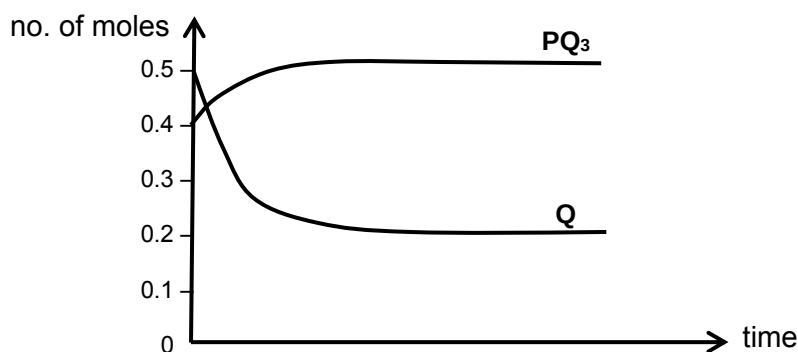
Which statement best explains the reason for the flattening off of the curve?

- A As the ethanol product builds up the reaction slows down.
- B All the ethanal has been used up and the reaction has finished.
- C At high ethanal concentrations the ethanal inhibits the action of the enzyme.
- D At high ethanal concentration all the active sites in the enzyme molecules are occupied by ethanal molecules.

- 14 The system containing P, Q and PQ_3 is allowed to reach equilibrium in a 10 dm^3 vessel at a temperature of 1000 K.



The diagram below shows the change in number of moles of PQ_3 and Q with time. The initial amount of P was 0.2 mol.



What is the equilibrium constant K_c for the reaction?

- A $\frac{0.5}{0.1 \times (0.2)^3}$ B $\frac{0.5}{0.2 \times (0.2)^3}$ C $\frac{0.5 \times 10^3}{0.1 \times (0.2)^3}$ D $\frac{0.5 \times 10^3}{0.2 \times (0.2)^3}$
- 15 Which statement is correct about a reaction for which the equilibrium constant is independent of temperature?
- 1 The rate constants for the forward and reverse reactions are also independent of temperature.
 - 2 Temperature has no effect on the position of equilibrium.
 - 3 The forward and reverse reactions have equal activation energies.
 - 4 There is no change in number of gaseous particles during reaction.
- A 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only
- 16 Which of the following is an **incorrect** descriptor of a weak acid?
- A It has high pK_a value.
B Its conjugate acid is strong.
C It donates a proton in aqueous solution.
D It has a relatively low electrical conductivity in dilute solutions.

[Turn over

- 17 50 cm³ of 0.10 mol dm⁻³ of solution **X** was added to 100 cm³ solution of 0.10 mol dm⁻³ HCO₃⁻ to form a buffer.

Which of the following could be the identity of **X**?

- 1 H₂CO₃
- 2 Na₂CO₃
- 3 H₂SO₄

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

- 18 A beaker contains a mixture of magnesium chloride, magnesium oxide, aluminium oxide and silicon dioxide. A student was asked to perform the following procedure to separate the compounds:

Step I: Add water to the mixture. Stir well and filter the mixture.
Step II: To the residue, add NaOH(aq) and stir well with a glass rod.
Step III: Filter the resulting mixture.
Step IV: To the residue, add HCl(aq) and stir well with a glass rod.
Step V: Filter to obtain the filtrate and residue.

Which of the following information is true about the above procedure?

- 1 The only starting species that remained unreacted after step V was silicon dioxide.
 - 2 After step I, the filtrate contains only magnesium chloride.
 - 3 After step III, the residue contains only aluminium oxide.
 - 4 A gas was produced in step IV.
- A 1 and 2 only
 - B 2 and 3 only
 - C 4 only
 - D None of the above

- 19 Which of the following is true about Period 3 elements and their compounds?
- A The covalent character of oxides decreases from sodium to silicon.
 - B The lattice energy of nitrates become less exothermic from sodium to aluminium.
 - C The solubility of oxides decreases from sodium to silicon.
 - D The melting points of the elements generally decrease across the Period.
- 20 The following information is about three Period 3 elements, X, Y and Z.
- The chloride of X reacts with water to give a colourless acidic solution.
 - The oxides of X and Y are insoluble in water.
 - The oxides of X and Y have high melting and boiling points whereas the oxide of Z has low melting and boiling points.

What is the order of increasing ionic radius for these three elements?

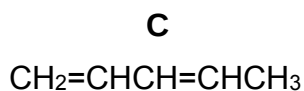
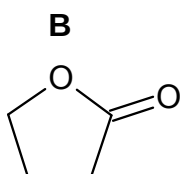
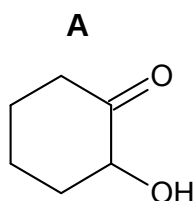
- A X, Y, Z
 - B Y, X, Z
 - C X, Z, Y
 - D Z, Y, X
- 21 Compound E has a straight-chain structure with a molecular formula of C_5H_8O .
- Which functional groups may be present in any of the isomers of compound E?

- 1 Alcohol
 - 2 Aldehyde
 - 3 Ester
- A 1, 2 and 3
 - B 1 and 2 only
 - C 1 only
 - D 2 only

[Turn over

- 22 Hydrogen gas was passed through the following compounds in the presence of platinum.

Which compound will produce a mixture of cis-trans isomers after the reaction?



- 23 Which compound does **not** have the same bond angle about all of its carbon atoms?

- A benzene
- B methanoic acid
- C 2-ethylpentan-2-ol
- D 2-bromopropene

- 24 $(\text{CH}_3\text{CH}_2)_3\text{CH}$ can react with chlorine under UV light to produce monochloro-compounds.

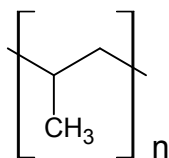
How many possible constitutional isomers of monochloro-compounds can it produce?

- A 2
- B 3
- C 6
- D 7

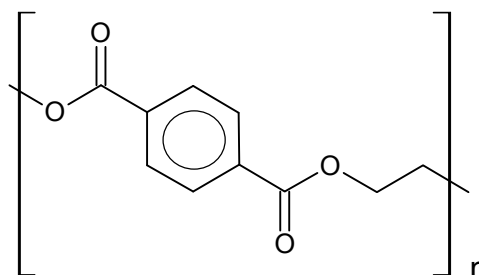
- 25 For which pair of compounds can ethanolic silver nitrate be used to distinguish between members of the pair?

- A $\text{CH}_3\text{CH}_2\text{Cl}$ and $\text{CH}_2=\text{CHCH}_2\text{Cl}$
- B $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3CHO
- C CH_3COCH_3 and CH_3CHO
- D $\text{CH}_3\text{CH}_2\text{Cl}$ and $\text{CH}_3\text{CH}_2\text{Br}$

- 26 Which compound will **not** produce a carboxylic acid when treated with hot acidified KMnO_4 ?
- A CH_3COCH_3
 B $\text{CH}_3\text{CH}_2\text{CHO}$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
 D $\text{CH}_3\text{CH}_2\text{CONHCH}_3$
- 27 Poly(propene) (PP) and poly(ethylene terephthalate) (PET) are frequently used as food grade containers.



PP



PET

Which statement is correct about their uses?

- A The presence of water will dissolve PET due to the formation of hydrogen bonds.
 B PP cannot be used to store oily foods as the non-polar PP will be dissolved by the non-polar oil molecules.
 C PP can be used to store strong alkalis but PET will be dissolved in strong alkalis due to alkaline hydrolysis of PET.
 D Both are unable to store hot foods due to their weak intermolecular forces of attraction that requires small amount of energy to break.
- 28 Which of the following are addition polymers?

- 1 Polyester
 2 Poly(ethene)
 3 Poly(vinyl alcohol)

- A 1 only
 B 2 and 3 only
 C 2 only
 D 1, 2 and 3

[Turn over

29 Carbon nanotubes are widely researched as a potential material with many uses.

Which pairs show the correct property of carbon nanotubes with an example of their use?

	Property	Example of use
1	low density	frame of racquets
2	high tensile strength	material for building construction
3	high electrical conductivity	flat screen television
4	high porosity	material for water filtration

- A** 2 and 3 only
- B** 1, 2, 3 and 4
- C** 1 and 2 only
- D** 1, 2 and 3 only

30 Which statement about nanomaterials is correct?

- A** They will not pose any health risks.
- B** They can have a length of more than 100 nm.
- C** They can only be manufactured in the laboratory.
- D** They are all able to stick onto any type of surfaces.